

Inverse Methods EAS 6134

Problem Set 2

1) We wish to show that the functional

$$N[\vec{x}] = \left(|x_1|^{1/2} + |x_2|^{1/2} + \dots + |x_n|^{1/2} \right)^2$$

is not a norm when the space is 2+ dimensions (i.e., $n > 1$).

A norm $\| \cdot \|$ must satisfy the following: (norm on V , let $f, g, h \in V$)

- i) $\|f\| \geq 0$ (positive-definite)
- ii) $\|\alpha f\| = |\alpha| \|f\|$ (linearity of scaling)
- iii) $\|f+g\| \leq \|f\| + \|g\|$ ($\| \cdot \|$ is sub-additive or concave)
- iv) $\|h\| = 0$ iff $h = 0$ (non-vanishing)

→ (i) is guaranteed since the absolute value is taken on the elements of $\vec{x}$ inside the radical, preventing complex values or negative norms

→ (ii) is satisfied by N ; let $n=k$, then

$$\begin{aligned} N[\vec{x}] &= \left(\sqrt{|x_1|} + \sqrt{|x_2|} + \dots + \sqrt{|x_k|} \right)^2 \\ \Rightarrow N[\alpha \vec{x}] &= \left(\sqrt{|\alpha x_1|} + \sqrt{|\alpha x_2|} + \dots + \sqrt{|\alpha x_k|} \right)^2 = \left(\sqrt{|\alpha|} (\sqrt{|x_1|} + \sqrt{|x_2|} + \dots + \sqrt{|x_k|}) \right)^2 \\ &= |\alpha| \left(\sqrt{|x_1|} + \dots + \sqrt{|x_k|} \right)^2 = |\alpha| N[\vec{x}] \end{aligned}$$

→ (iv) is automatically satisfied since $f(x) = \sqrt{x}$ and $f(x) = x^2$ are bijective on $\mathbb{R}^+$ (i.e., no nonzero values of x can possibly be mapped to zero through the norm)

→ However, property (iii) is not guaranteed, or even regularly satisfied by the functional N . We begin this demonstration by letting $n=2$.

Then,
$$N[\vec{x}] = \left(\sqrt{|x_1|} + \sqrt{|x_2|} \right)^2 = |x_1| + |x_2| + 2\sqrt{|x_1 x_2|} \geq N[x_1] + N[x_2],$$

where the equality is only for $x_1 = x_2 = 0$, i.e., the trivial case. Now, letting

$n=3$,
$$N[\vec{x}] = \left(\sqrt{|x_1|} + \sqrt{|x_2|} + \sqrt{|x_3|} \right)^2 = |x_1| + |x_2| + |x_3| + 2\sqrt{|x_1 x_2|} + 2\sqrt{|x_1 x_3|} + 2\sqrt{|x_2 x_3|}$$

via the binomial theorem. Hence, for $n=3$, $N[\vec{x}] \geq \sum_{i=1}^3 N[x_i]$. In general, it is clear that for $n=k$; $k \in \mathbb{N}$, $N[\vec{x}] = \left(\sqrt{|x_1|} + \sqrt{|x_2|} + \dots + \sqrt{|x_k|} \right)^2 = |x_1| + |x_2| + \dots + |x_k| + 2\sqrt{|x_1 x_2|} + \dots$

Written compactly,
$$N[\vec{x}] = |x_1| + \dots + |x_k| + 2\sqrt{|x_1 x_2|} + \dots + 2\sqrt{|x_1 x_k|} + \dots + 2\sqrt{|x_{k-1} x_k|}$$

$$\Rightarrow N[\vec{x}] = \sum_{i=1}^k |x_i| + 2 \sum_{l \neq m} \sqrt{|x_l x_m|}. \text{ Hence, since the term}$$

$\left(2 \sum_{l \neq m} \sqrt{|x_l x_m|} \right) > 0$ for any nonzero vector $\vec{x}$, ~~property 3/(iii) is not satisfied~~ by N . In other words, $N[\vec{x}] \geq \sum_{i=1}^k N[x_i]$.

Now that we have shown this, it will be straightforward to produce a counterexample that indicates condition (iii) is not always satisfied. To do so, let $\vec{f}, \vec{g} \in \mathbb{R}^{k+1}$, $\vec{x} \in \mathbb{R}^k$, and $g \in \mathbb{R}$. We define

$$\vec{f} \equiv \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ 0 \end{pmatrix} = \begin{pmatrix} \vec{x} \\ 0 \end{pmatrix} \quad \text{and} \quad \vec{g} \equiv \begin{pmatrix} 0 \\ \vdots \\ 0 \\ g \end{pmatrix} = \begin{pmatrix} \vec{0}_k \\ g \end{pmatrix}. \quad \text{Then,}$$

$$\begin{aligned} \|\vec{f} + \vec{g}\| &= N[\vec{f} + \vec{g}] = \left(\sqrt{|x_1|} + \sqrt{|x_2|} + \sqrt{|x_3|} + \dots + \sqrt{|x_k|} + \sqrt{|g|} \right)^2 \\ &= \sum_{i=1}^k |x_i| + |g| + 2 \sum_{l \neq m}^k \sqrt{|x_l x_m|} + 2 \sqrt{|x_1 g|} + 2 \sqrt{|x_2 g|} + \dots + 2 \sqrt{|x_k g|} \\ &= \sum_{i=1}^k |x_i| + |g| + 2 \sum_{l \neq m}^k \sqrt{|x_l x_m|} + 2 \sum_{j=1}^k \sqrt{|x_j g|} \\ &= \underbrace{\sum_{i=1}^k |x_i| + 2 \sum_{l \neq m}^k \sqrt{|x_l x_m|}}_{N[\vec{f}]} + \underbrace{|g|}_{N[\vec{g}]} + 2 \sum_{j=1}^k \sqrt{|x_j g|} \\ &= N[\vec{f}] + N[\vec{g}] + 2 \sum_{j=1}^k \sqrt{|x_j g|} \end{aligned}$$

$$\Rightarrow N[\vec{f} + \vec{g}] = N[\vec{f}] + N[\vec{g}] + 2 \sum_{j=1}^k \sqrt{|x_j g|}. \quad \text{Since } 2 \sum_{j=1}^k \sqrt{|x_j g|} > 0,$$

Condition (iii) is not satisfied by the functional N .

2) We wish to show that the trace

$$\text{Tr } \underline{A} = \sum_{j=1}^n A_{jj} = A_{ii} \text{ (Einstein)}$$

is an inner product on the space $S^{n \times n}$ of symmetric matrices. That is, if $\underline{A}, \underline{B} \in S^{n \times n}$, we wish to show that

$\text{Tr}(\underline{A} \cdot \underline{B}) = (\underline{A}, \underline{B})$ is an inner product on S . An inner product must satisfy:

i) $(\underline{A}, \underline{B}) = (\underline{B}, \underline{A})$ ii) $(\underline{A} + \underline{C}, \underline{B}) = (\underline{A}, \underline{B}) + (\underline{C}, \underline{B})$ iii) $(\alpha \underline{A}, \underline{B}) = \alpha (\underline{A}, \underline{B})$

iv) $(\underline{A}, \underline{A}) > 0$ if $\underline{A} \neq \underline{0}$

We will express the trace in Einstein notation to show (i)-(iv):

→ i) $(\underline{A}, \underline{B}) = \text{Tr}(\underline{A} \cdot \underline{B}) = A_{ij} B_{ji} = \sum_{j=1}^n \sum_{i=1}^n A_{ij} B_{ji} = \sum_{i=1}^n \sum_{j=1}^n B_{ji} A_{ij}$

If we let $i \leftrightarrow j$, then $\sum_{i=1}^n \sum_{j=1}^n B_{ji} A_{ij} = \sum_{j=1}^n \sum_{i=1}^n B_{ij} A_{ji} = \text{Tr}(\underline{B} \cdot \underline{A}) = (\underline{B}, \underline{A})$ ✓

→ ii) $(\underline{A} + \underline{C}, \underline{B}) = \text{Tr}((\underline{A} + \underline{C}) \cdot \underline{B}) = \text{Tr}(\underline{A} \cdot \underline{B} + \underline{C} \cdot \underline{B}) = A_{ij} B_{ji} + C_{ij} B_{ji} = \text{Tr} \underline{A} \cdot \underline{B} + \text{Tr} \underline{C} \cdot \underline{B} = (\underline{A}, \underline{B}) + (\underline{C}, \underline{B})$ ✓
(using distributivity of matrix multiplication)

→ iii) $(\alpha \underline{A}, \underline{B}) = \text{Tr}(\alpha \underline{A} \cdot \underline{B}) = (\alpha A_{ij}) B_{ji} = \alpha (A_{ij} B_{ji}) = \alpha \text{Tr}(\underline{A} \cdot \underline{B}) = \alpha (\underline{A}, \underline{B})$ ✓
← no sum index, factor out α

→ iv) $(\underline{A}, \underline{A}) = \text{Tr}(\underline{A} \cdot \underline{A}) = A_{ij} A_{ji} = \sum_j \left(\sum_i A_{ij} A_{ji} \right) = \sum_i A_{ii} A_{ii} + \sum_i A_{i2} A_{2i} + \dots + \sum_i A_{in} A_{ni} > 0$ ✓
unless zero or one element of $\underline{A}$ are non-vanishing

However, since $\underline{A} \in S^{n \times n}$, we have that

$$(\underline{A}, \underline{A}) = A_{ij} A_{ji} = A_{ii}^2 \text{ since } A_{ij} = A_{ji} \text{ by def.}$$

Hence, $(\underline{A}, \underline{A}) > 0$ unless $\underline{A} = \underline{0}$.

The norm associated with this inner product, given by $\sqrt{(\underline{A}, \underline{A})}$, can be written as the following:

$$(\underline{A}, \underline{A}) = \text{Tr}(\underline{A} \cdot \underline{A}) = A_{ij} A_{ji} \quad \text{(using } \underline{A} \in S^{n \times n} \text{)}$$

collapse index, interchange

Hence, the norm

$$\sqrt{(\underline{A}, \underline{A})} = \sqrt{A_{ij} A_{ji}} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2} = \sqrt{\sum_{i=1}^n \sum_{j=1}^n A_{ij}^2} = \|\underline{A}\|_F$$

(typo)

3) a.

#1	0.1
2	0.2
3	
4	
5	
...	

$$\underline{G} \cdot \underline{\vec{M}} = \underline{\vec{d}}$$

$\uparrow$ model operator $\uparrow$ model $\uparrow$ observations

Seismic velocity $\frac{v(z)}{dt} \rightarrow$ slowness $\frac{1}{s(z)} = \frac{dt}{dz} = \frac{1}{v(z)}$

observed travel times $\vec{d}$ given by

$$\vec{d} = d(z) = \int_0^z s(\xi) d\xi = \int_0^\infty s(\xi) H(z-\xi) d\xi$$

$$s(\xi) \approx M(\xi) \Rightarrow G(M) = \int_0^\infty M(\xi) H(z-\xi) d\xi$$

$$d = G(M)$$

$$d_i = G_i(M_i) = \int_0^\infty H(z_i - \xi) M(\xi) d\xi$$

$20 = z$ $\Delta x = 0.2$ $x_j = \frac{\Delta x}{2} + (j-1)\Delta x \equiv z_j$ ($z_j = [0.1M] + (j-1)[0.2M]$)

$$\Rightarrow d_i = \int_0^{x_i} s(\xi) d\xi = \int_0^\infty H(x_i - \xi) s(\xi) d\xi$$

$$d_i = d(z_i) = \int_0^\infty g_i(z) M(z) dz = \int_0^\infty H(z_i - \xi) M(\xi) d\xi$$

$$H(z_i - \xi) = g_i(\xi) \quad \text{"data kernel"}$$

$$\Rightarrow d_i = \int_0^\infty g_i(\xi) M(\xi) d\xi \approx \sum_{j=1}^N g_i(z_j) M(z_j) \Delta x$$

or, more simply

$$d_i = \sum_{j=1}^i M(z_j) \Delta x = \sum_{j=1}^i s(z_j) \Delta x$$

without introducing arbitrary complexity for no sake.

Hence, $\vec{d} = \underline{G} \cdot \vec{s}$ where

$$\underline{G} = \begin{pmatrix} \Delta x & 0 & 0 & \dots & 0 \\ \Delta x & \Delta x & 0 & 0 & \dots & 0 \\ \Delta x & \Delta x & \Delta x & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \Delta x & \Delta x & \dots & \dots & \dots & \Delta x \end{pmatrix}$$

(lower triangular)

$$G_{ij} = \begin{cases} \Delta x \delta_{ij} & i \geq j \\ 0 & i < j \end{cases}$$

, i.e.,

3) (continued)

b. slowness $s(z) = \frac{1}{v(z)}$

Let $v(z) = v_0 + kz$

$\{v_0 = 1 \text{ m/s}; k = 40 \text{ s}^{-1}\}$

$\Rightarrow \vec{M}_{true} = \vec{s}(\vec{z}) \Rightarrow \frac{1}{v_0 + kz_i} = s_i$

$$d(z) = \int_0^z s(\xi) d\xi = \int_0^z \frac{1}{v_0 + k\xi} d\xi$$

$$= \frac{1}{k} \log(v_0 + k\xi) \Big|_0^z$$

$$= \frac{1}{k} \log(v_0 + kz) - \frac{1}{k} \log(v_0)$$

$$= \frac{1}{k} \log\left(\frac{v_0 + kz}{v_0}\right) = \frac{1}{k} \log\left(1 + \frac{kz}{v_0}\right)$$

for i th index:

$$\Rightarrow \begin{cases} d_i = \frac{1}{k} \log\left(1 + \frac{kz_i}{v_0}\right) \\ M_i = s_i = \frac{1}{v_0 + kz_i} \end{cases}$$

3) [continued]

c. See attached figure

d. See attached Figure(s). A copy of the Figure was included demonstrating that the apparent increase of noise with depth is due only to the choice of g -axis scaling (i.e., log vs. linear). The erratic noise can be seen to slightly influence the obtained slowness values, but the general behavior remains accurate.

e. See attached Figure. The $n=4$ sensor depths are calculated analogously to part (a), where no sensor is present at $z=0$ and the spacing $\Delta z = \frac{20 \text{ m}}{4} = 5 \text{ m}$ depths.

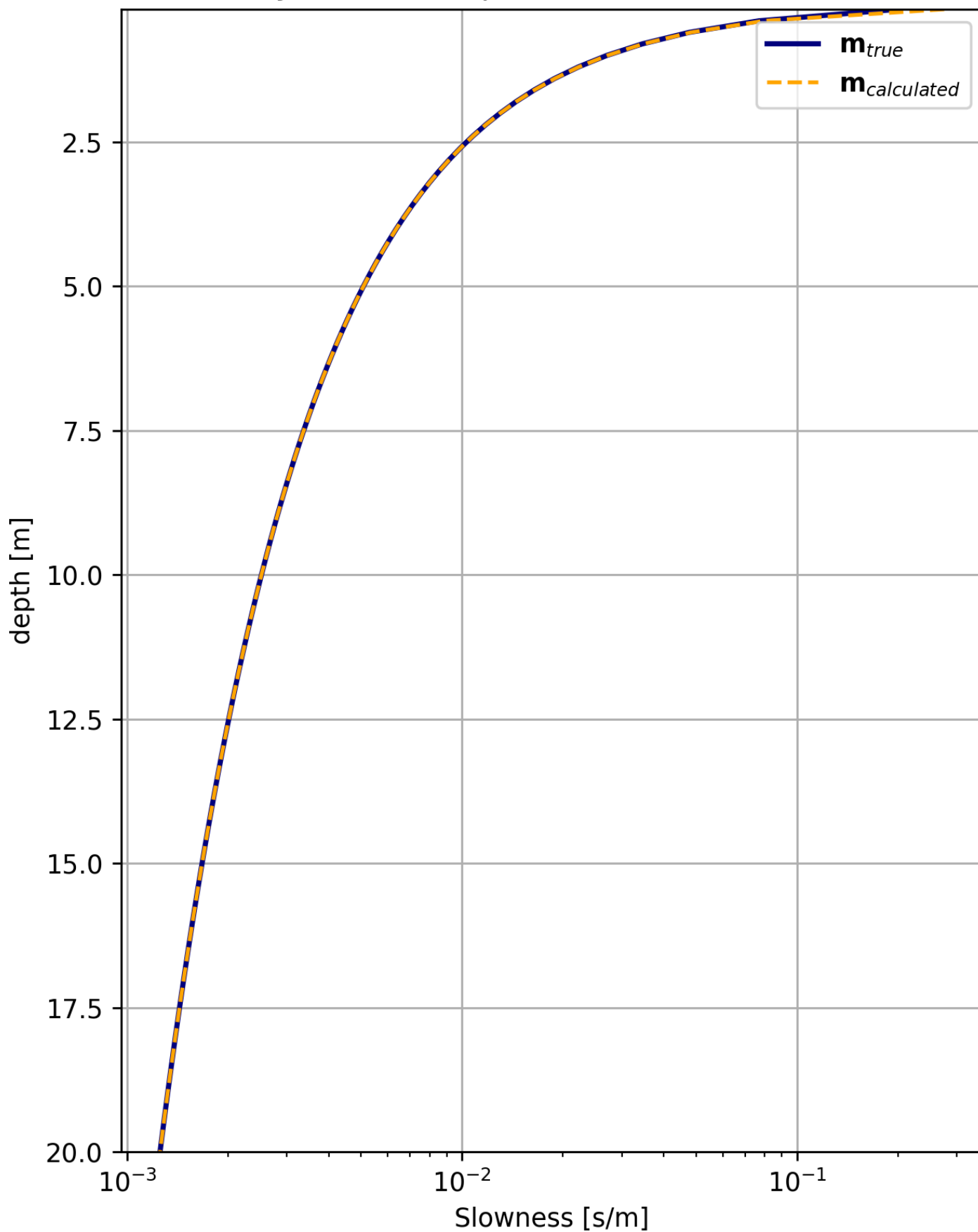
The condition number of our operators $\underline{G}_{n=4}$ and $\underline{G}_{n=100}$ represent their sensitivity to changes in input; or, if an input includes error, how that error propagates via the mapping $\underline{G}$ is set by the condition number. We can see that the fit, especially at lower depths, is weaker for the $n=4$ case. The condition numbers, and the propagation of error, however, are both apparently different between the cases as well:

$$\text{for } n=4, \quad \text{cond}(\underline{G}) \approx 5.4$$

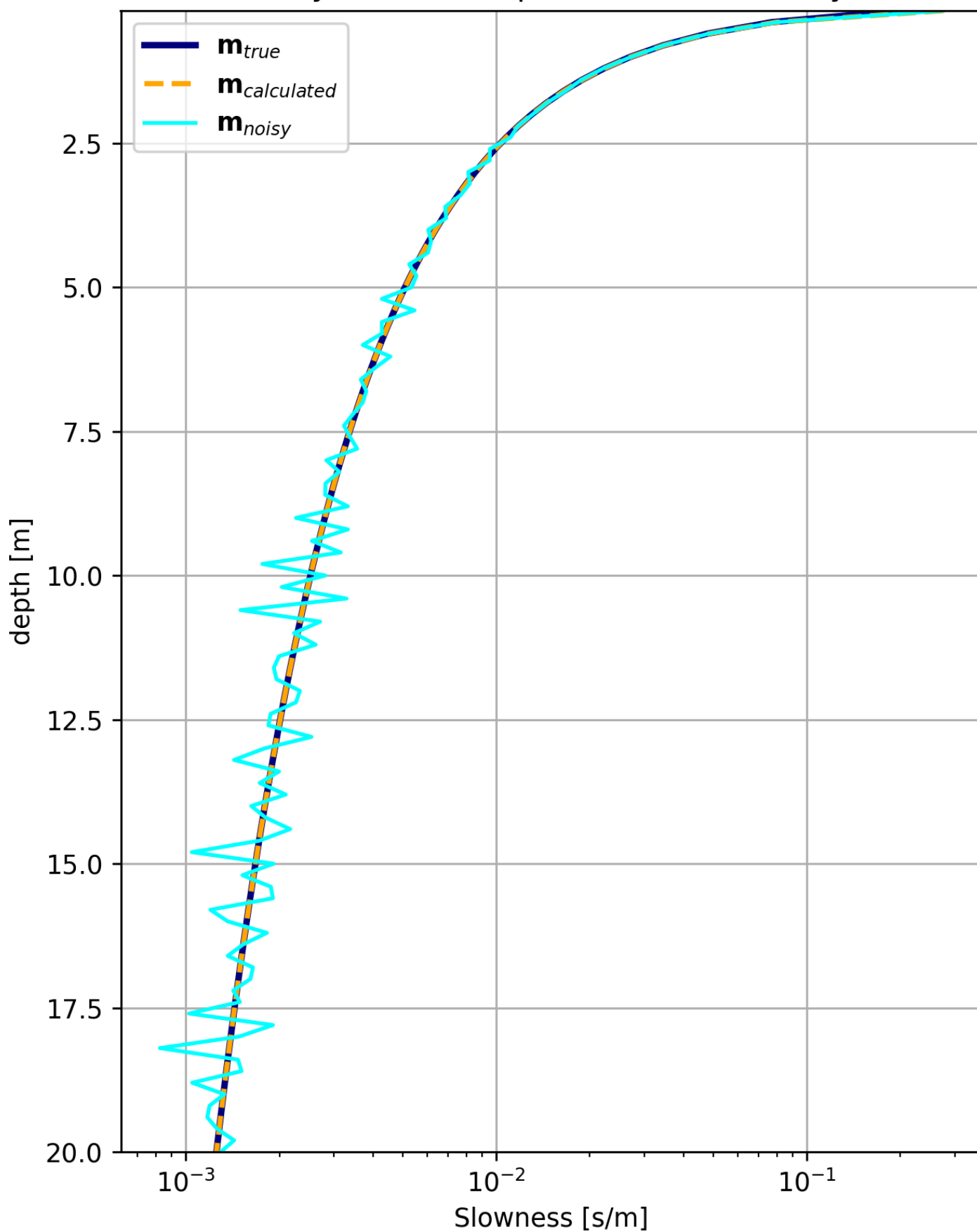
$$\text{for } n=100, \quad \text{cond}(\underline{G}) \approx 127.9$$

Hence, we expect the error introduced by the prescribed Gaussian noise to affect the $n=100$ sensor array data more, even though the integration method itself is improved. This is reflected directly by our data (synthetic), where the coarse array inversion produces a worse fit at lower depths, but the error has almost no effect. In contrast, the fit for $n=100$ is substantially impacted by noise and the calculated inversions with/without noise differ much more, on average, than for the $n=4$ case.

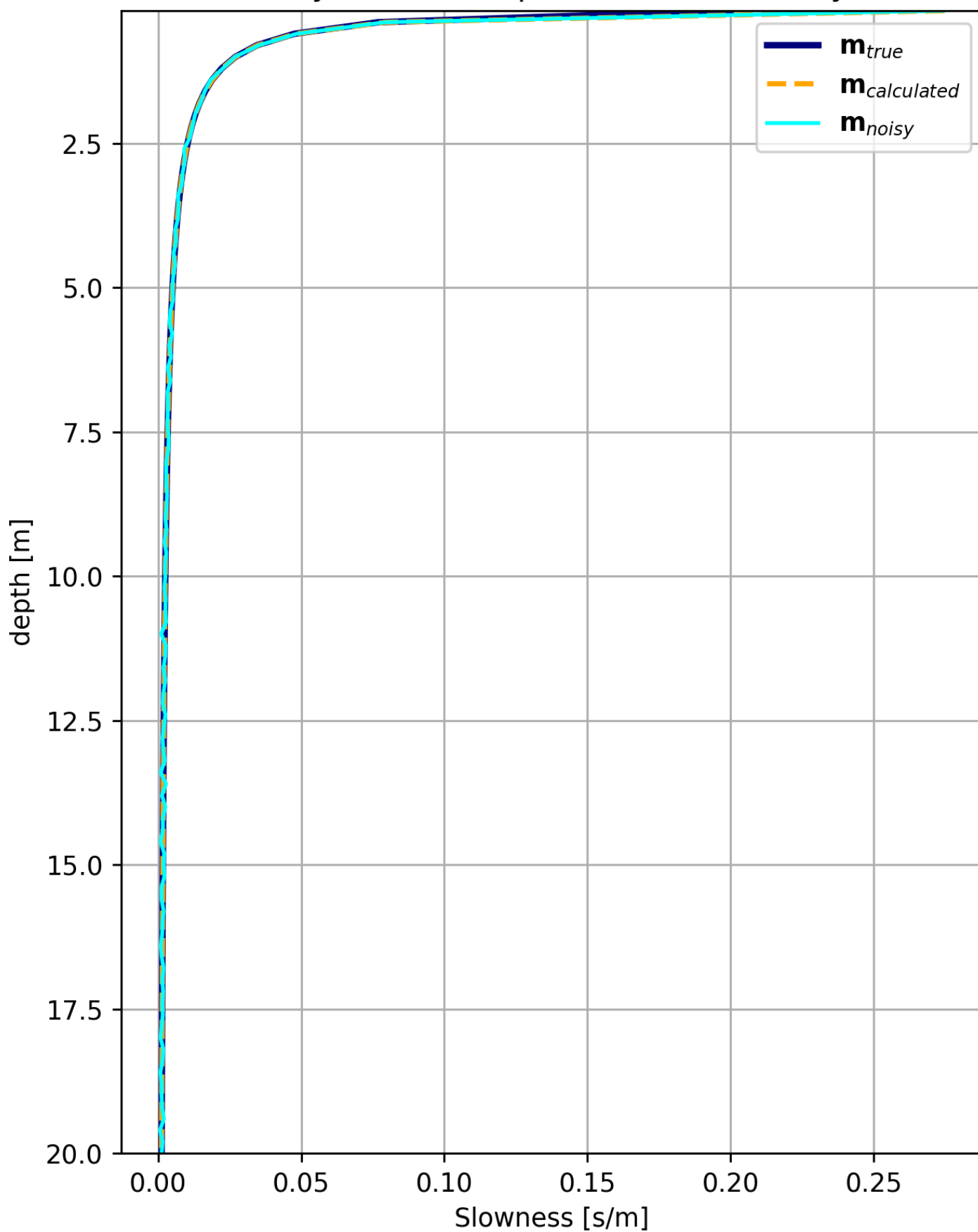
Slowness vs. Depth:
Analytical vs. Simple Collocation (noise-free)



Slowness vs. Depth:
Analytical vs. Simple Collocation (noisy)



Slowness vs. Depth:
Analytical vs. Simple Collocation (noisy)



Slowness vs. Depth:
Analytical vs. Simple Collocation (coarse sensor array)

